

Vijay Siga

Data Engineer

Email: Sigavijay28@gmail.com [Linkedin](https://www.linkedin.com/in/vijay-s-b0a1b3367/): <https://www.linkedin.com/in/vijay-s-b0a1b3367/> Mobile: 214-899-0697 Location: Frisco, TX

SUMMARY

Versatile Data Engineer with extensive experience in designing and implementing **ETL/ELT pipelines, data lakes, and real-time streaming solutions** across **Azure, AWS, and Hadoop ecosystems**. Proficient in **Databricks, PySpark, SQL, Airflow, Snowflake, DB2, and Informatica**, with strong expertise in handling structured and semi-structured datasets for enterprise reporting and analytics. Skilled in building **secure, governed, and scalable data platforms** using **Delta Lake, Unity Catalog, and cloud-native services**. Adept at **workflow orchestration** with **Airflow and Control-M**, as well as **data visualization** using **Power BI, Tableau, and Python frameworks**. Strong background in **data governance, CI/CD automation, and cloud integration**, ensuring reliable, high-quality data delivery for business stakeholders.

PROFESSIONAL EXPERIENCE

APPWORKS

TEXAS, USA

Data Engineer

March 2024 – Present

- Designed and automated scalable ETL pipelines using **Databricks, PySpark, and SQL** for batch and streaming ingestion from 50 U.S. states; enhanced automation and performance with **Linux shell scripting**.
- I have expertise in **SSIS, SSRS, SQL, Power BI, and Informatica**, with experience across **Azure, AWS, and DataStage**, delivering scalable **ETL/BI** solutions and proven success in data governance and architecture design.
- Migrated legacy systems to **AWS (S3, Glue, Athena)** and **Azure (ADLS Gen2, ADF, Synapse)** data lakes, improving query performance by 45%.
- Implemented real-time data streaming with Azure Event Hubs (Kafka)** and Databricks Structured Streaming, enabling **low-latency ingestion (<2 sec delay)** for enterprise analytics dashboards..
- Built and managed **Delta Lake tables** with ACID compliance and Hive compatibility for secure, governed SQL-based access.
- Optimized **Spark jobs** using broadcast joins, caching, and partitioning, reducing runtime by 35%.
- Orchestrated **batch and streaming workflows** by integrating ADF with Databricks and leveraging **Control-M** for enterprise-scale scheduling and monitoring.
- Implemented **Delta Live Tables with Auto Loader** for continuous schema-evolving ingestion with minimal operational overhead.
- Developed **interactive dashboards** using **Python (Matplotlib, Plotly)**, Tableau, and Power BI to present demographic and population insights to stakeholders.
- Designed and optimized **complex SQL queries and ETL logic** for large-scale demographic and geospatial datasets, improving processing by 30%.
- Performed **daily data validation and quality checks** using Spark SQL in Databricks to ensure accurate reporting datasets.
- Enabled **Unity Catalog** for fine-grained access control, lineage tracking, and compliance with governance standards.
- Designed and developed **Snowflake-based data pipelines**, implementing features like **time travel, zero-copy cloning, and micro-partitioning** to optimize storage and query performance for analytical workloads
- Built scalable ETL pipelines in Databricks using PySpark and Delta Lake**, automating ingestion from 50 U.S. state data sources — achieving **35% faster processing** and improving data reliability across AWS S3 and Azure ADLS Gen2.
- Orchestrated data workflows using Airflow and Azure Data Factory** integrated with Control-M, achieving **99.9% SLA compliance** and automated lineage tracking with Unity Catalog.

Cleveland State University

Cleveland, OH

Data Engineer

Nov 2022 – Nov 2023

- Developed and optimized Informatica-based ETL pipelines for Oracle data warehouses, integrating with Power BI and Python for analytics — resulting in 40% reduced data refresh times.
- Modeled large datasets on Oracle systems to ensure scalability, consistency, and data integrity for analytics.
- Built **Power BI dashboards** to deliver actionable insights by transforming and modeling Oracle data for business stakeholders.
- Automated ETL orchestration with Apache Airflow DAGs** and Python validation scripts (pandas, SQLAlchemy), improving **data quality by 25%** and ensuring full job monitoring and error recovery.

- **Modeled relational and semi-structured datasets (JSON, XML)** for enterprise reporting, enhancing **data consistency and scalability** across academic and operational systems.
- Implemented **SQL scripts with stored procedures, triggers, and indexing** to optimize Oracle database performance for large-scale reporting.
- Orchestrated ETL jobs using **Apache Airflow**, creating DAGs for scheduling, monitoring, and error handling of critical pipelines.
- Integrated **Python and Airflow** to perform automated data validation, quality checks, and notifications for failed tasks.

JOHNSON CONTROLS INC.

HYDERABAD, INDIA

Data Engineer

Apr 2018 – Oct 2022

- **Designed end-to-end ETL/ELT pipelines using Azure Databricks, PySpark, and Synapse Analytics**, improving query performance and dashboard refresh by **40%**. Automated workflows with **Control-M**, ensuring SLA-driven execution and reliability.
- Engineered **CDC-enabled Delta Lake tables** with schema evolution and time travel for real-time analytics and historical tracking.
- Implemented **multi-hop Delta Lake architecture (Bronze → Silver → Gold)** with schema enforcement and Auto Loader for ingestion.
- Built **executive dashboards** in Power BI and Tableau with row-level security, DAX, and parameterized views for secure real-time reporting.
- Modeled datasets in **Azure Synapse** with T-SQL, optimized using materialized views and distribution strategies, reducing dashboard load times by 40%.
- Created Hadoop-based applications and automation scripts with **Python and Pandas**, improving data loading efficiency by 15%.
- Implemented **Unity Catalog, Azure AD passthrough, and Key Vault-backed credentials** for secure and governed data access.
- **Migrated data warehouse from IBM DB2 to Snowflake and Redshift**, implementing CDC and schema evolution with Delta Lake for **real-time analytics** and **zero data loss**.
- **Implemented CI/CD for Databricks jobs via Azure DevOps and GitHub**, ensuring **standardized, automated deployments** and reducing manual intervention by **60%**.

TECHNICAL SKILLS

Languages	: Python, PySpark, Shell Scripting, Bash
ETL Tools	: Informatica PowerCenter, Informatica IICS, Talend, dbt (Data Build Tool), Control-M
Data Warehouses	: Snowflake, Amazon Redshift, Oracle, Azure Synapse Analytics
Big Data Tools	: Apache Spark (Core, SQL, Streaming), Hadoop, HDFS, Hive
Database Management	: Oracle PL/SQL, MS SQL Server, PostgreSQL, MySQL, MongoDB, Google BigQuery
Cloud Provider & Services	: Azure Cloud Services, Databricks, Azure Data Lake, ADF, AWS, Azure Synapse, S3, Athena, Glue.
Testing & Version Control	: ETL Testing, Git, GitHub, Bitbucket, Kafka, SVN, Control-M
Data Visualization Tools	: MS Office, Power BI, Advance Excel, Tableau, Looker, AWS Quick sight

CERTIFICATIONS

Azure Fundamentals AZ-900
 DP -900
 Advanced Google Analytics Certification.

EDUCATION

Master of Science in Information Science
 Bachelor of Technology in Electronics & Communication

Cleveland State University, Ohio, USA
 Guru Nanak Institute of Technical Campus, India